OPTIMAL O(N^(1/4)) DISCREPANCY FOR PLANAR TRIANGLES

GLEB SMIRNOV

Editorial note. Complete original mathematical paper109–801 and original support/diagrams retained. Main input17 is expanded as int.def2–627 preamble spans. Helper line1 loads unused epsf.tex; neither main nor helper uses EPS commands, so only that unused loader is omitted in rendering. Raw main/helper unchanged.

Editorial note. Complete original147-line main.bbl inserted at bibliography position, with original ref.bib retained. No bibliographic metadata reconstructed. Original proof style and every triangle TikZ coordinate/label preserved.

Editorial note. Original main.tex173–174 credits ChatGPT for suggesting a simpler proof of Proposition slabs (original proof used OU process), assistance with estimates in prop:halfplanes, and editing. Literal disclosure and AI_SCOPE.json retained. Only thm:triangles selected; unselected slabs/halfplanes not counted. No AI-free label, whole-paper origin inference or AI-dependency investigation.

Editorial note. One principal object; prefix/slab support, known lower bounds, rotated-rectangle consequences and cited predecessor results not additional counts. Exact original CC BY4.0 retained.

Editorial note. Author owns mathematics. No proof repair, new hypotheses, mathematical review, counterexample search, numerical/symbolic replay, content hashing, version comparison or certification.

Editorial note. Actual r001 missing t2aenc.def: rendering uses existing T1 for the original Latin text/bibliography; no Cyrillic text occurs in the native main/bbl. Original raw T2A declaration and failed r001 preserved. Mathematical macros, assumptions, proof and native diagrams unchanged.

Editorial note. Actual final log reports U+2013 missing in ec-lmr10. The original bibliography contains five en-dash page ranges; rendering copy uses equivalent TeX – for those five page ranges only. Raw main.bbl/ref.bib and all actual bibliographic values unchanged.

Allowed standard prerequisites (editorial).

- Original discrepancy, Fisher-information, halfplane-packing and quoted partial-coloring setup. All native definitions, domain/parameter restrictions and cited background theorems retained.. Complete author slab-to-prefix-to-triangle proof chain supplied without external-proof reconstruction or certification.

Date: arxiv:2610.00375v1; CC BY4.0.

1. Main results. Let $X = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$ be a set of points. We color each point either $+1$ or -1. We want every triangle to contain nearly as many $+1$'s as -1's. To make this precise, encode a coloring by a vector $\chi \in \{-1,1\}^n$: the point p_i gets color χ_i. For a subset $F \subset X$, write:

$$\chi(F) := \sum_{p_i \in F} \chi_i.$$

We want $|\chi(X \cap T)|$ to be small for all triangles T at once.

More generally, let $\mathcal{R}$ be a family of subsets of $\mathbb{R}^2$. The *discrepancy* of X wrt $\mathcal{R}$ is the imbalance of the best coloring:

$$\operatorname{disc}(X, \mathcal{R}) := \min_{\chi \in \{-1,1\}^n} \sup_{R \in \mathcal{R}} |\chi(X \cap R)|.$$

Our main result is the following statement.

Theorem 1.1. *Every set $X = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$ has a coloring $\chi \in \{-1,1\}^n$ such that:*

$$|\chi(X \cap T)| \leqslant C\, n^{1/4} \qquad \text{for all triangles } T,$$

where $C > 0$ is an absolute constant.

For planar triangles, the best bound we are aware of is $O(n^{1/4}\sqrt{\log n})$. This bound follows from more general results of Matoušek et al. [24]. See Chazelle [14, § 1.4] for a simple exposition.

Theorem 1.1 is sharp. The lower bound is a classical result of Beck [11]. A rectangle is a union of two triangles. Hence, it suffices to prove the lower bound for rectangles in arbitrary position. Chazelle [14, § 3.2] and Beck [11] explain the following lower bound for their Lebesgue discrepancy. For every n-point set $X \subset [0,1]^2$, there is a rotated rectangle $R \subset [0,1]^2$ such that:

$$\big|\#(X \cap R) - n \operatorname{area}(R)\big| = \Omega(n^{1/4}).$$

Beck's transference principle [9] carries this bound over to the combinatorial discrepancy.

We deduce Theorem 1.1 from a statement about halfplanes. A *halfplane trace* is a set $X \cap P$, where $P \subset \mathbb{R}^2$ is a halfplane. Let $\mathcal{H}(X)$ denote the family of halfplane traces. Let the points of X be ordered. For $0 \leqslant k \leqslant n$, set:

$$X_k := \{p_1, \dots, p_k\}.$$

Theorem 1.2. *Every ordered $X = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$ has a coloring $\chi \in \{-1,1\}^n$ such that:*

$$|\chi(F \cap X_k)| \leqslant C\, n^{1/4} \qquad \forall F \in \mathcal{H}(X), \quad 0 \leqslant k \leqslant n,$$

where $C > 0$ is an absolute constant.

For $k = n$, this is Matoušek's bound for halfplanes [22]. The point is that one coloring controls all prefixes at once. We explain the step from halfplanes to triangles in § 5.

We prove Theorem 1.2 using the Fisher information technique of [28, 29]. The halfplane constraints cut out a convex body in $\mathbb{R}^n$; it is an intersection of slabs. The crucial step is to bound the Fisher information of densities supported on intersections of slabs. We derive such bounds in § 2; they may be of independent interest. The full proof occupies § 2, § 3, and § 4.

We now discuss other related work.

Theorem 1.2 is a vector balancing statement. Indeed, for each p_i, let $u_i \in \{0,1\}^{\mathcal{H}(X)}$ be the indicator of the halfplane traces that contain p_i. Then Theorem 1.2 balances all prefix sums $\sum_{i \leqslant k} \chi_i u_i$ in ℓ_∞.

Vector balancing has seen rapid progress. For the Komlós problem, Bansal and Jiang [7] improved Banaszczyk's $O(\log^{1/2} n)$ bound to $O(\log^{1/4} n)$. Guo, Fang and Lu [17] obtained a constant bound; this settles the Komlós conjecture. See the work of Karingula and Lovett [18] and of Akbas and Sra [2, 3] for alternative proofs. In the online Komlós problem, the vectors arrive one by one, and all prefix sums must stay small. Optimal bounds are obtained by Kulkarni, Reis and Rothvoss [19] and by Aden-Ali [1]. Their work improves earlier bounds of Alweiss, Liu and Sawhney [5] and Bansal et al. [8]. The online Spencer problem, where the vectors are only bounded in ℓ_∞, was solved by Altschuler and Tikhomirov [4].

Our proof relies on the partial coloring method. The method goes back to Beck [10] and Spencer [30]. Constructive methods for partial coloring are due to Bansal [6] and Lovett and Meka [20]. Our mechanism for partial coloring is different from all these. These methods need the convex body to be large: in entropy or in Gaussian measure. We need instead a density supported in it with small Fisher information.

For discrepancy of geometric set systems, combinatorial and Lebesgue, see the monographs [12, 14, 21] and references therein. For halfplanes, the bound $O(n^{1/4})$ is due to Matoušek [22]. Similar to that work, our proof relies on chaining and uses the packing lemma for halfplanes; see Mustafa [25] for a simple proof. For axis-parallel boxes, see Matoušek et al. [23] and Nikolov [26]. On the Lebesgue discrepancy, see also Chen and Travaglini [16] for convex polygons and Bilyk and Mastrianni [13] for rectangles with restricted rotations. By Beck's transference principle [9], an upper bound on combinatorial discrepancy gives a corresponding upper bound for Lebesgue discrepancy. In particular, Theorem 1.1 gives $O(n^{1/4})$ for the Lebesgue discrepancy of rotated rectangles. This improves Beck's bound $O(n^{1/4}\sqrt{\log n})$ [12], and matches his lower bound.

Acknowledgment and AI assistance. The author thanks OpenAI's ChatGPT for useful insights: ChatGPT suggested a simpler proof of Proposition 2.1, while the original proof relied on the OU process. ChatGPT also helped the author with estimates in the proof of Proposition 3.2. Finally, ChatGPT helped to edit the manuscript.

2. Fisher cost of intersecting slabs. Let $K \subset \mathbb{R}^n$ be a convex body. We seek a function $\psi \in C^\infty(\mathbb{R}^n)$ with

$$\|\psi\|_{L^2(\mathbb{R}^n)} = 1, \qquad \operatorname{supp}\psi \subset K,$$

and with small

$$\int_{\mathbb{R}^n} \|\nabla\psi\|_2^2.$$

This quantity controls the Metropolis rejections below. In [28], it equals $\operatorname{tr} M_\psi$. The bodies K we need are intersections of slabs. Here, we estimate $\int \|\nabla\psi\|_2^2$ for such intersections.

To begin with, we introduce a cutoff function. Fix an even function $g \in C^\infty(\mathbb{R})$, non-increasing on $[0,\infty)$, such that:

$$g(t) = \begin{cases} 1, & |t| \leqslant 1/2, \\ > 0, & |t| < 1, \\ 0, & |t| \geqslant 1, \end{cases}$$

and such that $\log g$ is *concave* on $(-1,1)$. In particular, $0 \leqslant g \leqslant 1$. Set:

$$U := -2\log g \quad \text{on } (-1,1).$$

Then U is convex, and $\ddot{U} = 0$ on $(-1/2, 1/2)$. From

$$g^2\ddot{U} = 2\dot{g}^2 - 2g\ddot{g},$$

we get:

$$\sup_{|t|<1} g(t)^2\,\ddot{U}(t) \leqslant C, \tag{2.1}$$

for some C depending only on g.

Proposition 2.1. *Let $v_1,\dots,v_M \in \mathbb{R}^n$ and $a_1,\dots,a_M > 0$. Set:*

$$K := \big\{z \in \mathbb{R}^n :\ |\langle v_j, z\rangle| \leqslant a_j \quad \forall j\big\},$$

and assume K is bounded. Define:

$$\psi(z) := Z^{-1/2}\, e^{-\|z\|_2^2/4} \prod_{j=1}^{M} g\Big(\frac{\langle v_j, z\rangle}{a_j}\Big),$$

where Z is chosen so that $\|\psi\|_{L^2(\mathbb{R}^n)} = 1$. Then:

$$4\int_{\mathbb{R}^n} \|\nabla\psi\|_2^2 \leqslant n + 2\,C \sum_{j=1}^{M} \frac{\|v_j\|_2^2}{a_j^2} \exp\Big(-\frac{a_j^2}{8\|v_j\|_2^2}\Big). \tag{2.2}$$

Proof. Since $0 \leqslant g \leqslant 1$,

$$Z\psi(z)^2 \leqslant e^{-\|z\|_2^2/2},$$

with equality near $z = 0$. Hence, $0 < Z < \infty$. Clearly, $\psi \in C^\infty(\mathbb{R}^n)$ and $\operatorname{supp}\psi \subset K$. Since K is bounded, $\psi \in C_c^\infty(\mathbb{R}^n)$.

Let $\Omega := \{\psi > 0\}$; this is the interior of K. Write:

$$\psi^2 = Z^{-1} e^{-V}, \quad \text{where} \quad V(z) = \frac{\|z\|_2^2}{2} + \sum_{j=1}^{M} U\Big(\frac{\langle v_j, z\rangle}{a_j}\Big).$$

Then:

$$\Delta V = n + \sum_{j=1}^{M} \frac{\|v_j\|_2^2}{a_j^2}\, \ddot{U}\Big(\frac{\langle v_j, z\rangle}{a_j}\Big) \quad \text{on } \Omega. \tag{2.3}$$

Also, $V = -2\log\psi - \log Z$ on Ω. Hence:

$$\psi^2 \Delta V = 2\|\nabla\psi\|_2^2 - 2\psi\Delta\psi \quad \text{on } \Omega.$$

Consequently,

$$\int_\Omega \psi^2 \Delta V = \int_\Omega \big(2\|\nabla\psi\|_2^2 - 2\psi\Delta\psi\big).$$

ψ vanishes outside Ω, and so does $\nabla\psi$. Therefore:

$$\int_\Omega \psi^2 \Delta V = \int_{\mathbb{R}^n} \big(2\|\nabla\psi\|_2^2 - 2\psi\Delta\psi\big).$$

Since $\psi \in C_c^\infty(\mathbb{R}^n)$, integration by parts gives:

$$-\int_{\mathbb{R}^n} \psi\Delta\psi = \int_{\mathbb{R}^n} \|\nabla\psi\|_2^2.$$

Consequently,

$$\int_\Omega \psi^2 \Delta V = 4\int_{\mathbb{R}^n} \|\nabla\psi\|_2^2.$$

Let us substitute (2.3) for ΔV. By (2.1), each $\ddot{U}(\langle v_j, z\rangle/a_j)\,\psi^2$ is at most C/Z; and Ω is bounded. So each term is integrable, and we obtain:

$$4\int_{\mathbb{R}^n} \|\nabla\psi\|_2^2 = n + \sum_{j=1}^{M} \frac{\|v_j\|_2^2}{a_j^2} \int_\Omega \ddot{U}\Big(\frac{\langle v_j, z\rangle}{a_j}\Big)\psi(z)^2\,dz. \tag{2.4}$$

Let us bound each integral on the right. Fix j. For each z, use the orthogonal decomposition:

$$z = y\,\frac{v_j}{\|v_j\|_2} + w, \quad y = \frac{\langle z, v_j\rangle}{\|v_j\|_2}, \quad w \perp v_j.$$

Then:

$$\psi(z)^2 = \frac{1}{Z}\,e^{-y^2/2}\,g\Big(\frac{\|v_j\|_2\,y}{a_j}\Big)^2 \cdot e^{-\|w\|_2^2/2} \prod_{k\neq j} g\left(\frac{y\left\langle v_k, \frac{v_j}{\|v_j\|_2}\right\rangle + \langle v_k, w\rangle}{a_k}\right)^2.$$

Let $z \sim \psi^2$. Integrating over $w \in v_j^\perp$, we find the density of y:

$$p_j(y) = \frac{1}{Z}\,e^{-y^2/2}\,g\Big(\frac{\|v_j\|_2\,y}{a_j}\Big)^2 \int_{v_j^\perp} e^{-\|w\|_2^2/2} \prod_{k\neq j} g\left(\frac{y\left\langle v_k, \frac{v_j}{\|v_j\|_2}\right\rangle + \langle v_k, w\rangle}{a_k}\right)^2 dw.$$

Set:

$$H(y,w) := e^{-\|w\|_2^2/2} \prod_{k\neq j} g\left(\frac{y\left\langle v_k, \frac{v_j}{\|v_j\|_2}\right\rangle + \langle v_k, w\rangle}{a_k}\right)^2, \qquad h(y) := \int_{v_j^\perp} H(y,w)\,dw.$$

Then:

$$p_j(y) = \frac{1}{Z}\,e^{-y^2/2}\,g\Big(\frac{\|v_j\|_2\,y}{a_j}\Big)^2 h(y).$$

g is log-concave on $\mathbb{R}$; hence, each factor of H is log-concave in the joint variable (y,w). Thus, $H(y,w)$ is log-concave. By Prékopa's theorem (eg, [27]), $h(y)$ is also log-concave.

g is even; hence, $H(y,w) = H(-y,-w)$. Thus:

$$h(y) = \int_{v_j^\perp} H(y,w)\,dw = \int_{v_j^\perp} H(-y,-w)\,dw = \int_{v_j^\perp} H(-y,w')\,dw' = h(-y),$$

and $h(y)$ is also even.

An even log-concave function is non-increasing on $[0,\infty)$. Therefore:

$$\forall s,t \geqslant 0 \qquad h(s+t) \leqslant h(s). \tag{2.5}$$

Let us bound the j-th integral in (2.4). We compute:

$$\int_\Omega \ddot{U}\Big(\frac{\langle v_j, z\rangle}{a_j}\Big)\psi^2 = \mathbb{E}_y\,\ddot{U}\Big(\frac{\|v_j\|_2\,y}{a_j}\Big) = \frac{1}{Z}\int_{\mathbb{R}} (g^2\ddot{U})\Big(\frac{\|v_j\|_2\,y}{a_j}\Big)e^{-y^2/2}h(y)\,dy,$$

where the integrand is set to zero for $|y| \geqslant a_j/\|v_j\|_2$. By (2.1), $g^2\ddot{U} \leqslant C$. Also, $\ddot{U} = 0$ on $(-1/2, 1/2)$. Hence:

$$\int_\Omega \ddot{U}\Big(\frac{\langle v_j, z\rangle}{a_j}\Big)\psi^2 \leqslant C\int_{t<|y|<2t} q(y)\,dy, \qquad \text{where } \; t := \frac{a_j}{2\|v_j\|_2}, \quad q(y) := \frac{1}{Z}\,e^{-y^2/2}h(y).$$

From (2.5):

$$\forall s \geqslant 0 \quad q(s+t) = \frac{1}{Z}\, e^{-(s+t)^2/2} h(s+t) \leqslant \frac{1}{Z}\, e^{-s^2/2} e^{-t^2/2} h(s) = e^{-t^2/2} q(s).$$

Hence:

$$\int_t^{2t} q(y)\, dy \leqslant e^{-t^2/2} \int_0^t q(y)\, dy.$$

q is even; hence:

$$\int_{t<|y|<2t} q(y)\, dy = 2 \int_{t<y<2t} q(y)\, dy.$$

Therefore:

$$\int_\Omega \ddot{U}\Big(\frac{\langle v_j, z\rangle}{a_j}\Big)\psi^2 \leqslant 2Ce^{-t^2/2} \int_0^t q(y)\, dy.$$

For $|y| \leqslant t$, $g(\|v_j\|_2\, y/a_j) = 1$; hence: $q(y) = p_j(y)$. Thus:

$$\int_0^t q(y)\, dy = \int_0^t p_j(y)\, dy = \mathbb{P}\big(0 \leqslant y \leqslant t\big) \leqslant 1.$$

Consequently,

$$\int_\Omega \ddot{U}\Big(\frac{\langle v_j, z\rangle}{a_j}\Big)\psi^2 \leqslant 2Ce^{-t^2/2} = 2C \exp\Big(-\frac{a_j^2}{8\|v_j\|_2^2}\Big),$$

and the proof follows. □

3. Chaining. Let $X = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$. A *halfplane trace* is a set $X \cap P$, where $P \subset \mathbb{R}^2$ is a halfplane. Let $\mathcal{H}(X)$ denote the family of halfplane traces. For $F \subset X$, let $\mathbf{1}_F \in \mathbb{R}^n$ be its indicator vector:

$$(\mathbf{1}_F)_i = \begin{cases} 1, & \text{if } p_i \in F, \\ 0, & \text{if } p_i \notin F. \end{cases}$$

Thus:

$$\langle \mathbf{1}_F, z\rangle = \sum_{p_i \in F} z_i \quad \text{for } z \in \mathbb{R}^n.$$

For $D > 0$, set:

$$K_D(X) := \big\{z \in \mathbb{R}^n : \ |\langle \mathbf{1}_F, z\rangle| \leqslant D \text{ for all } F \in \mathcal{H}(X)\big\}.$$

$K_D(X)$ is convex, bounded, and centrally symmetric. It is also an intersection of slabs. However, we do not apply Proposition 2.1 to these slabs directly. Instead, we write each $\mathbf{1}_F$ as a sum of short increments and impose slabs on the increments.

We use the following known packing result. See [21, § 5.3] and [15].

Theorem 3.1. *For $\delta > 0$, a family $\mathcal{F} \subset \mathcal{H}(X)$ is* δ-separated *if $|F \triangle F'| > \delta$ for all distinct $F, F' \in \mathcal{F}$. There is an absolute constant C_0 such that for every $0 < \delta \leqslant n$ and every δ-separated $\mathcal{F} \subset \mathcal{H}(X)$:*

$$|\mathcal{F}| \leqslant C_0\, (n/\delta)^2.$$

For $\delta < 1$, Theorem 3.1 is just the bound $|\mathcal{H}(X)| = O(n^2)$.

Nets. Let us use Theorem 3.1 to build a hierarchy of nets in $\mathcal{H}(X)$. This greedy construction is well known. For $F, G \in \mathcal{H}(X)$, set:

$$d(F,G) := |F \triangle G|.$$

Then:

$$\|\mathbf{1}_F - \mathbf{1}_G\|_2^2 = d(F,G).$$

Set:

$$L := \lceil \log_2 n \rceil + 1, \qquad \delta_i := n2^{-i} \qquad 0 \leqslant i \leqslant L.$$

Set $\mathcal{F}_0 := \{\varnothing\}$. For $1 \leqslant i \leqslant L$, let $\mathcal{F}_i \subset \mathcal{H}(X)$ be a maximal δ_i-separated family. By Theorem 3.1, $|\mathcal{F}_i| \leqslant C_0 4^i$.

By maximality, every $G \in \mathcal{H}(X)$ satisfies $d(G,F) \leqslant \delta_i$ for some $F \in \mathcal{F}_i$. Hence, each $F \in \mathcal{F}_i$, $i \geqslant 2$, has a *parent* $p(F) \in \mathcal{F}_{i-1}$ with $d(F,p(F)) \leqslant \delta_{i-1}$. For $F \in \mathcal{F}_1$, set $p(F) := \varnothing$; then $d(F,p(F)) = |F| \leqslant \delta_0$. In all cases:

$$d(F,p(F)) \leqslant s_i := n2^{1-i} \qquad \forall F \in \mathcal{F}_i.$$

Also, $\delta_L \leqslant 1/2$. Since d is integer-valued, $\mathcal{F}_L = \mathcal{H}(X)$.

Increments. For $1 \leqslant i \leqslant L$, set:

$$\mathcal{V}_i := \left\{\mathbf{1}_F - \mathbf{1}_{p(F)} : \ F \in \mathcal{F}_i\right\} - \{0\}.$$

Fix $H \in \mathcal{H}(X)$. Since $\mathcal{F}_L = \mathcal{H}(X)$, we may set $F_L := H$. Then set:

$$F_{i-1} := p(F_i) \quad \text{for } i = L, \dots, 1.$$

The increments telescope:

$$\mathbf{1}_H = \sum_{i=1}^{L} \left(\mathbf{1}_{F_i} - \mathbf{1}_{F_{i-1}}\right). \tag{3.1}$$

Along with (3.1), below we use the following properties of the increments:

$$|\mathcal{V}_i| \leqslant C_0 4^i, \qquad \|v\|_2^2 \leqslant s_i \quad \forall v \in \mathcal{V}_i. \tag{3.2}$$

Proposition 3.2. *For each $D > 0$, there exists $\psi_D \in C_c^\infty(\mathbb{R}^n)$ such that:*

$$\|\psi_D\|_{L^2(\mathbb{R}^n)} = 1, \quad \operatorname{supp} \psi_D \subset K_D(X),$$

and such that:

$$\int_{\mathbb{R}^n} \|\nabla \psi_D\|_2^2 \leqslant C_1 \frac{n^{3/2}}{D^2}, \tag{3.3}$$

where $C_1 > 0$ is an absolute constant.

Proof. Choose k_0 so that:

$$\sqrt{n} \leqslant 2^{k_0} < 2\sqrt{n}.$$

One may take $k_0 := \lceil \frac{1}{2} \log_2 n \rceil$. Set:

$$b_i := n^{1/4} 2^{-|i-k_0|/4},$$

and set:

$$K' := \left\{z \in \mathbb{R}^n : \ |\langle v, z\rangle| \leqslant b_i \text{ for all } v \in \mathcal{V}_i \text{ and all } i\right\}.$$

By (3.1), for $z \in K'$ and $H \in \mathcal{H}(X)$:

$$|\langle \mathbf{1}_H, z\rangle| \leqslant \sum_{i=1}^{L} |\langle \mathbf{1}_{F_i} - \mathbf{1}_{F_{i-1}}, z\rangle| \leqslant \sum_{i=1}^{L} b_i,$$

since each $\mathbf{1}_{F_i} - \mathbf{1}_{F_{i-1}}$ is either 0 or in $\mathcal{V}_i$. Consequently,

$$|\langle \mathbf{1}_H, z\rangle| \leqslant Bn^{1/4}, \qquad B := \sum_{i\in\mathbb{Z}} 2^{-|i|/4}.$$

Hence:

$$K' \subset K_{Bn^{1/4}}(X). \tag{3.4}$$

In particular, K' is bounded.

Fisher cost. Apply Proposition 2.1 to K': the vectors are $v \in \mathcal{V}_i$, and the width of $v \in \mathcal{V}_i$ is b_i. Let ψ be the resulting function. By (2.2),

$$\int_{\mathbb{R}^n} \|\nabla\psi\|_2^2 \leqslant n + O\Big(\sum_{i=1}^{L} \sum_{v\in\mathcal{V}_i} f_i\big(\|v\|_2^2\big)\Big), \qquad \text{where} \quad f_i(x) := \frac{x}{b_i^2} \exp\Big(-\frac{b_i^2}{8x}\Big).$$

Each f_i is increasing on $(0,\infty)$. By (3.2), $\|v\|_2^2 \leqslant s_i$ for $v \in \mathcal{V}_i$; hence, $f_i(\|v\|_2^2) \leqslant f_i(s_i)$. Also by (3.2), the inner sum has at most $C_0 4^i$ terms. Therefore:

$$\int_{\mathbb{R}^n} \|\nabla\psi\|_2^2 \leqslant n + O\Big(\sum_{i=1}^{L} 4^i \frac{s_i}{b_i^2} \exp\Big(-\frac{b_i^2}{8s_i}\Big)\Big).$$

Let us bound the i-th term. Write:

$$j := i - k_0.$$

Since $2^{k_0} < 2\sqrt{n}$,

$$4^i = 4^j 4^{k_0} = O(n\,4^j).$$

Since $2^{k_0} \geqslant \sqrt{n}$,

$$s_i = n2^{1-i} = 2n2^{-k_0}\,2^{-j} \leqslant 2\sqrt{n}\,2^{-j}.$$

Since $b_i^2 = \sqrt{n}\,2^{-|j|/2}$,

$$4^i \frac{s_i}{b_i^2} = O\big(n\,2^{j+|j|/2}\big), \qquad \frac{b_i^2}{8s_i} \geqslant \frac{2^{j-|j|/2}}{16}.$$

Case $j \leqslant 0$. In this case,

$$4^i \frac{s_i}{b_i^2} = O\big(n\,2^{j+|j|/2}\big) = O\big(n\,2^{-|j|/2}\big).$$

We bound the i-th term as follows:

$$4^i \frac{s_i}{b_i^2} \exp\Big(-\frac{b_i^2}{8s_i}\Big) \leqslant 4^i \frac{s_i}{b_i^2} = O\big(n\,2^{-|j|/2}\big).$$

Case $j > 0$. In this case,

$$4^i \frac{s_i}{b_i^2} = O\big(n\,2^{j+|j|/2}\big) = O\big(n\,2^{3j/2}\big), \qquad \frac{b_i^2}{8s_i} \geqslant \frac{2^{j-|j|/2}}{16} = \frac{2^{j/2}}{16}.$$

We bound the i-th term as follows:

$$4^i \frac{s_i}{b_i^2} \exp\Big(-\frac{b_i^2}{8s_i}\Big) \leqslant O\big(n\,2^{3j/2}\big) \cdot \exp\big(-\frac{2^{j/2}}{16}\big) = O\big(n\,2^{3j/2}\big) \cdot O(2^{-2j}) = O\big(n\,2^{-j/2}\big).$$

In both cases, the i-th term is $O(n\,2^{-|j|/2})$. Therefore:

$$\sum_{i=1}^{L} 4^i \,\frac{s_i}{b_i^2}\exp\Big(-\frac{b_i^2}{8s_i}\Big) \leqslant O\Big(\sum_{j\in\mathbb{Z}} n\,2^{-|j|/2}\Big) = O(n).$$

Consequently,

$$\int_{\mathbb{R}^n} \|\nabla\psi\|_2^2 \leqslant n + O\Big(\sum_{i=1}^{L} 4^i\,\frac{s_i}{b_i^2}\exp\Big(-\frac{b_i^2}{8s_i}\Big)\Big) = n + O(n) = O(n).$$

Scaling. Now, we rescale ψ. Set:

$$r := D/(Bn^{1/4}), \quad \psi_D(z) := r^{-n/2}\psi(z/r).$$

Then $\|\psi_D\|_{L^2(\mathbb{R}^n)} = 1$. Also, $\operatorname{supp}\psi_D = r\operatorname{supp}\psi$. By (3.4),

$$\operatorname{supp}\psi_D \subset rK_{Bn^{1/4}}(X) = K_D(X).$$

Finally,

$$\int \|\nabla\psi_D\|_2^2 = r^{-2}\int \|\nabla\psi\|_2^2, \quad \int_{\mathbb{R}^n} \|\nabla\psi_D\|_2^2 = O\big(r^{-2}n\big) = O\Big(\frac{n^{3/2}}{D^2}\Big),$$

and the proof follows. □

4. Prefix halfplanes. Recall the setting of § 3. Let again $X = \{p_1,\ldots,p_n\} \subset \mathbb{R}^2$ be a set of points, $\mathcal{H}(X)$ be the family of halfplane traces. For a subset $F \subset X$, let $\mathbf{1}_F \in \mathbb{R}^n$ be its indicator vector. Set again:

$$K_D(X) = \big\{z \in \mathbb{R}^n :\ |\langle \mathbf{1}_F, z\rangle| \leqslant D \text{ for all } F \in \mathcal{H}(X)\big\}.$$

Now, let $X = \{p_1,\ldots,p_n\} \subset \mathbb{R}^2$ be an *ordered* set of points. For $0 \leqslant k \leqslant n$, set:

$$X_k := \{p_1,\ldots,p_k\}.$$

For $\chi \in \mathbb{R}^n$ and $F \subset X$, write:

$$\chi(F) := \langle \mathbf{1}_F, \chi\rangle = \sum_{p_i\in F} \chi_i.$$

Theorem 4.1. *Every $X = \{p_1,\ldots,p_n\}$ has a coloring $\chi \in \{-1,1\}^n$ with:*

$$|\chi(F \cap X_k)| = O(n^{1/4}) \qquad \forall F \in \mathcal{H}(X),\quad 0 \leqslant k \leqslant n.$$

The proof relies on the following result of [28].

Theorem 4.2. *Let $K \subset \mathbb{R}^n$ be convex and centrally symmetric. Let $\psi \in C_c^\infty(\mathbb{R}^n)$ be such that:*

$$\|\psi\|_{L^2(\mathbb{R}^n)} = 1, \qquad \operatorname{supp}\psi \subset K.$$

Let $v_1,\ldots,v_m \in \mathbb{R}^n$ be any vectors, and let $\varepsilon_1,\ldots,\varepsilon_m$ be iid fair signs. Then there is an online algorithm that discards some terms $\varepsilon_i v_i$ so that, for each k:

$$\sum_{i\leqslant k \text{ accepted}} \varepsilon_i v_i \in 2K \quad \textit{deterministically,}$$

and:

$$\mathbb{E}\,\#\{\text{discarded}\} \leqslant \sum_{i=1}^{m} \big\|\langle\nabla\psi, v_i\rangle\big\|_{L^2(\mathbb{R}^n)}.$$

First, we color most of the points.

Lemma 4.3. *Every $X = \{p_1, \ldots, p_n\}$ has a partial coloring $\chi \in \{-1, 0, 1\}^n$ with at most $n/4$ zeros and with:*

$$|\chi(F \cap X_k)| = O(n^{1/4}) \qquad \forall F \in \mathcal{H}(X), \quad 0 \leqslant k \leqslant n.$$

Proof. Fix $D > 0$, and choose it later. Apply Theorem 4.2 to $K = K_D(X)$: choose $\psi = \psi_D$ from Proposition 3.2, and $v_i := e_i$, $i = 1, \ldots, n$. Set:

$$\chi_i := \begin{cases} \varepsilon_i, & \text{if } \varepsilon_i e_i \text{ is accepted,} \\ 0, & \text{if } \varepsilon_i e_i \text{ is discarded.} \end{cases}$$

Then:

$$\forall k \quad \sum_{i \leqslant k} \chi_i e_i \in 2K_D(X).$$

Consequently,

$$|\chi(F \cap X_k)| \leqslant 2D \qquad \forall F \in \mathcal{H}(X), \quad 0 \leqslant k \leqslant n. \tag{4.1}$$

By Theorem 4.2,

$$\mathbb{E}\, \#\{\chi_i = 0\} \leqslant \sum_{i=1}^{n} \|\partial_i \psi_D\|_{L^2(\mathbb{R}^n)}.$$

By Cauchy-Schwarz,

$$\mathbb{E}\, \#\{\chi_i = 0\} \leqslant \sqrt{n} \Big(\int_{\mathbb{R}^n} \|\nabla \psi_D\|_2^2 \Big)^{1/2}.$$

From (3.3):

$$\mathbb{E}\, \#\{\chi_i = 0\} = O\Big(\frac{n^{5/4}}{D}\Big).$$

Choose $D = O(n^{1/4})$ so that $\mathbb{E}\, \#\{\chi_i = 0\} \leqslant n/4$. Then there exists some χ with at most $n/4$ zeros, and the proof follows. □

Proof of Theorem 4.1. The following iteration is standard; see [30].

We iterate Lemma 4.3, as in [29, § 2]. First, apply it to X. Then apply it to the points left uncolored, in the induced order, and so on until all points are colored. Round r is applied to at most $n4^{-r}$ points. A halfplane trace of X, restricted to these points, is a halfplane trace of them. A prefix of X, restricted to these points, is a prefix of them. Let χ be the final coloring. Then $\chi(F \cap X_k)$ splits into one term per round. By Lemma 4.3, the term from round r is $O\big((n4^{-r})^{1/4}\big)$; therefore:

$$|\chi(F \cap X_k)| \leqslant \sum_{r \geqslant 0} O\big((n4^{-r})^{1/4}\big) = O(n^{1/4}),$$

and the proof follows. □

5. Triangles. Let $X = \{p_1, \dots, p_n\} \subset \mathbb{R}^2$ be a set of points. For $\chi \in \mathbb{R}^n$ and $F \subset X$, write again:

$$\chi(F) = \sum_{p_i \in F} \chi_i.$$

Theorem 5.1. *Every $X = \{p_1, \dots, p_n\}$ has a coloring $\chi \in \{-1, 1\}^n$ with:*

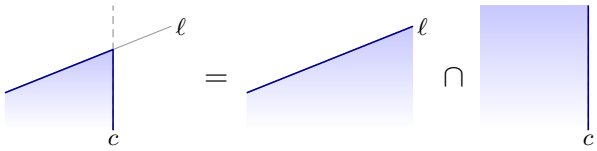
$$|\chi(X \cap T)| = O(n^{1/4}) \qquad \textit{for all triangles } T.$$

Proof. The proof reduces triangles to corners. For a nonvertical line $\ell\colon y = \ell(x)$ and $c \in \mathbb{R}$, set:

$$\Gamma(\ell, c) := \big\{(x, y) \in \mathbb{R}^2 :\ x \leqslant c,\ \ y \leqslant \ell(x)\big\}.$$

Order X by the first coordinate. Let χ be the coloring from Theorem 4.1 for this order.

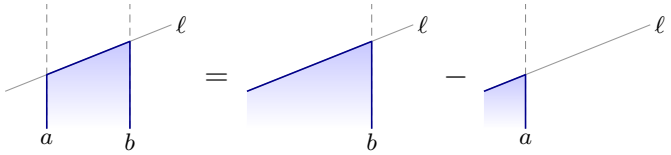
Corners. $X \cap \Gamma(\ell, c)$ is a halfplane trace intersected with a prefix:



By Theorem 4.1,

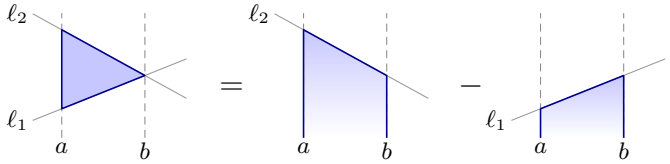
$$|\chi(X \cap \Gamma(\ell, c))| = O(n^{1/4}). \tag{5.1}$$

Interfixes. Consider the domain below ℓ and between the vertical lines $x = a$ and $x = b$; it is a difference of two corners:



Its trace is a halfplane trace intersected with an interfix $X_k - X_j$. By (5.1), its χ is $O(n^{1/4})$.

Triangles with a vertical side. Such a triangle is a difference of two domains of the previous kind:



For such a triangle, its χ is again $O(n^{1/4})$.

Triangles. Every triangle T is a union of two triangles with vertical sides. Perturbing T slightly, we may assume that no point of X lies on the boundaries in the pictures. Hence, $\chi(X \cap T)$ is the sum of two terms, each $O(n^{1/4})$. □

References.

[1] I. Aden-Ali. Optimal online discrepancy minimization in linear time, 2026.
[2] E. Akbas and S. Sra. Boolean small-ball inequalities for discrepancy theory, 2026.
[3] E. Akbas and S. Sra. An exposition of the proof of the Komlós conjecture, 2026.
[4] D. Altschuler and K. Tikhomirov. Online Beck-Fiala down to logarithmic sparsity, 2026.
[5] R. Alweiss, Y. Liu, and M. Sawhney. Discrepancy minimization via a self-balancing walk. In *STOC 2021*, pages 14–20, 2021.
[6] N. Bansal. Constructive algorithms for discrepancy minimization. In *FOCS*, pages 3–10, 2010.

[7] N. Bansal and H. Jiang. Decoupling via affine spectral-independence: Beck-Fiala and Komlós bounds beyond Banaszczyk. 2025.
[8] N. Bansal, H. Jiang, R. Meka, S. Singla, and M. Sinha. Online discrepancy minimization for stochastic arrivals. In *SODA 2021*, pages 2842–2861, 2021.
[9] J. Beck. Balanced two-colorings of finite sets in the square I. *Combinatorica*, 1(4):327–335, 1981.
[10] J. Beck. Roth's estimate of the discrepancy of integer sequences is nearly sharp. *Combinatorica*, 1(4):319–325, 1981.
[11] J. Beck. Irregularities of distribution. I. *Acta Math.*, 159:1–49, 1987.
[12] J. Beck and W. Chen. *Irregularities of Distribution*. Cambridge University Press, 1987.
[13] D. Bilyk and M. Mastrianni. Lower bounds for the directional discrepancy with respect to an interval of rotations. *Journal of Fourier Analysis and Applications*, 29(3), 2023.
[14] B. Chazelle. *The Discrepancy Method*. Cambridge University Press, 2000.
[15] B. Chazelle and E. Welzl. Quasi-optimal range searching in spaces of finite VC-dimension. *Discr. Comp. Geom.*, 4(5):467–489, 1989.
[16] W. Chen and G. Travaglini. Discrepancy with respect to convex polygons. *J. Complexity*, 23:662–672, 2007.
[17] S Guo, E. Fang, and J. Lu. Vector balancing via directional total variation, 2026.
[18] S. Karingula and S. Lovett. An elementary proof of the Komlós conjecture, 2026.
[19] J. Kulkarni, V. Reis, and T. Rothvoss. Optimal online discrepancy minimization. In *STOC 2024*, page 1832–1840, 2024.
[20] S. Lovett and R. Meka. Constructive discrepancy minimization by walking on the edges. *SIAM J. Comput.*, 44:1573–1582, 2015.
[21] J. Matoušek. *Geometric Discrepancy*. 1999.
[22] J. Matoušek. Tight upper bounds for the discrepancy of half-spaces. *Discr. Comp. Geom.*, 13:593–601, 1995.
[23] J. Matoušek, A. Nikolov, and K. Talwar. Factorization norms and hereditary discrepancy. *IMRN*, (3):751–780, 2020.
[24] J. Matoušek, E. Welzl, and L. Wernisch. Discrepancy and approximations for bounded VC-dimension. *Combinatorica*, 13:455–466, 1993.
[25] N. Mustafa. A simple proof of the shallow packing lemma. *Discr. Comp. Geom.*, 55:739–743, 2016.
[26] A. Nikolov. Tighter bounds for the discrepancy of boxes and polytopes. *Mathematika*, 63:1091–1113, 2017.
[27] A. Saumard and J. Wellner. Log-concavity and strong log-concavity: a review. *Statistics Surveys*, 8:45–114, 2014.
[28] G. Smirnov and R. Vershynin. Discrepancy and Fisher information. 2026.
[29] G. Smirnov and R. Vershynin. Random sets are close to low-discrepancy sets. 2026.
[30] J. Spencer. Six standard deviations suffice. *Tran. Amer. Math. Soc.*, 289(2):679–706, 1985.